

Margaret Yan

Wellesley College
21 Wellesley College Road
Wellesley, MA 02481

(201) 678-8602
my106@wellesley.edu
<https://maggiesyan.framer.website>

EDUCATION

Wellesley College, Wellesley, MA

Expected May 2027

B.A. in Media Arts & Sciences, Minor in Economics | Cumulative GPA: 3.83/4.0

Massachusetts Institute of Technology (MIT), Cambridge, MA

Jan 2025 - Present

Cross-registered student | Cumulative GPA: 5.0/5.0

RESEARCH INTERESTS

Human-computer interaction and human-AI interaction, with interests in AI trust, co-creation, and usability in health and decision-making contexts. Areas of focus include trust, transparency, and adoption of AI-assisted decision support across clinician- and patient-facing systems; explainable AI (XAI); human AI co-creation and collaborative design; mental health technology design; human-centered design for health technology and medical devices; accessible and inclusive design in research and maker environments; human understanding of and response to anthropomorphic systems.

RESEARCH EXPERIENCE

Massachusetts Institute of Technology (MIT), Cambridge, MA

Sep 2026 - Present

MIT Media Lab, Responsive Environments Group

Undergraduate Research Assistant

- Co-design and run in-lab user studies that expands a pilot study of a wearable silent-speech recognition device that uses face- and neck-mounted piezoelectric sensors, managing participant recruitment, scheduling, consent, and hardware and recording software during data collection sessions.
- Contribute to data labeling, cleaning, and statistical analysis of collected sessions, applying machine learning methods, including connectionist temporal classification (CTC) loss, to model silent-speech signals.
- Document study protocols and help draft manuscript sections for a planned submission to ACM UIST 2027.

Massachusetts Institute of Technology (MIT), Cambridge, MA

Sep 2026 - Present

MIT Media Lab, Critical Matter Group

Undergraduate Research Assistant

- Ran user studies that evaluated how people respond to novel end-effector systems developed for a UR20 robotic arm as part of an interdisciplinary robotics and materials research project, and produced the supplementary video for its ACM CHI 2027 paper submission; translated research into a concise visual format for the review process.
- Contribute to the prototyping and fabrication of a large-scale art installation featuring live electrospinning demonstrations for MIT's Open House celebrating the opening of the new School of Architecture building.

Washington University in St. Louis, St. Louis, MO

Aug 2026 - Present

Institute for Informatics, Data Science & Biostatistics, Abraham Lab

Visiting Researcher

- Screen titles and abstracts for a scoping review of EHR-derived indicators and measure of clinical care team teamwork, using a structured eligibility criteria for care setting, data source, and level of team measurement.

Washington University in St. Louis, St. Louis, MO

May 2026 - July 2026

Institute for Informatics, Data Science & Biostatistics, Abraham Lab

Bioinformatics Research Intern

- Collected observational and interview data on the OR-to-ICU handoff process across orthopedic, cardiovascular/cardiothoracic, and neurology units through in-person case observations, handwritten field notes, and clinician conversations encouraging the use of an Epic-embedded handoff report.
- Analyzed field notes and clinician feedback using inductive thematic analysis and open coding, from initial codes through sub-themes to themes, to identify handoff workflow challenges and perceived benefits and barriers of the Epic report.
- Contributed to mapping the existing OR-to-ICU handoff workflow from these observational findings to identify opportunities for embedding a machine learning (ML)-based risk prediction tool.
- Collected usability data for the third phase of a mixed-methods study evaluating a ML risk prediction model embedded in the Epic handoff tool by participating in Zoom-based cognitive walkthrough sessions with OR and ICU clinicians.
- Analyzed session recordings using a mix of inductive and deductive coding, then synthesized findings across all three study phases with the research team to identify where the tool would be most valuable within the handoff workflow.

Massachusetts Institute of Technology (MIT), Cambridge, MA

Aug 2025 - Aug 2026

Institute for Medical Engineering and Science, Roche Lab

Undergraduate Research Assistant

- Develop interactive patient dashboards in Plotly Dash to centralize and analyze multimodal trial data across sleep sensor outputs (Nox Medical T3, Noxturnal), suction pump metrics, and qualitative pre- and post-trial survey responses.
- Contribute to device development for the ZzAlign sleep apnea device by processing patient dental scans and customizing personalized 3D-printed mouthguard designs in Autodesk Fusion to prepare models for fabrication.
- Design 3D medical animations in Blender that demonstrates the function of the device for research presentations and potential investor pitches.
- Support participant recruitment for ongoing clinical trials by developing creative recruitment materials and coordinating study outreach through social media.
- Design and iteratively fabricate a silicone tongue, casted from 3D-printed molds modeled in Shapr3D, for a benchtop otolaryngological anatomical model used to test and refine mouthguard designs.
- Assist with thermoforming mouthguards as a part of the fabrication process, which would be used directly in testing for the ongoing clinical study.

Wellesley College Clapp Library, Wellesley, MA

Apr 2025 - Present

Knapp Makerspace & Interactive Media Lab

Research Intern: Accessibility & Human-Centered Design

- Conduct a human-centered design research project to assess and improve makerspace accessibility, focusing on spatial design, equipment usability, and instructional inclusivity.
- Performed a narrative literature review on accessibility in makerspaces; analyzed research papers and case studies to identify key barriers and best practices.

- Conducted user interviews with administrative staff and student organizations focused on campus accessibility, to gather insights and consult on actionable improvements.
- Reviewed the Massachusetts Accessibility Code (521 CMR) and federal ADA standards along with Knapp's space; concluded that an accessible space required going beyond a baseline compliance.
- Led a site visit with staff from Wellesley College's Office of Accessibility and Disability Resources (ADR) to identify accessibility gaps, implemented three interventions based on ADR feedback, which included adjustable lighting, a sensory support station with face masks and ear plugs, and signage using more inclusive language; evaluated the impact via a follow-up survey.
- Co-organized a community workshop with a student organization, Students for an Accessible Wellesley (SAW), to learn directly from students about accessibility barriers, then developed the framework for an interactive, visual novel-style training guide for makerspace student workers, prototyped in Figma.

Massachusetts Institute of Technology (MIT), Cambridge, MA

Feb 2025 - May 2025

MIT Media Lab, Fluid Interfaces Group

Undergraduate Research Assistant

- Designed and refined survey questions for the NeuraFutures research project to gather data on public understanding of and emotional responses to brain-computer interfaces (BCIs), paying attention to clarity and bias reduction of survey questions.
- Pilot-tested and iteratively revised the survey to improve clarity and data quality across diverse respondent populations, strengthening the overall study design.
- Developed a lifelike robotic head prototype modeled after the InMoov Head i2, including silicone skin molding, mechanical repairs, and electrical testing, to be used in a study protocol exploring human reactions to the uncanny valley.

PUBLICATIONS

Yan, M., Wang, W., Regan, E.

Framing Financial Feedback: Narrative vs. Factual AI Explanation in an Educational Simulation

Extended abstract accepted to ACM SIGDOC 2026, Student Research Competition; to appear in the conference proceedings after October 24, 2026. <https://doi.org/10.1145/3772063.3841690>

Extended Abstract:  [yan-regan-wang-sigdoc-finalvers.pdf](#) | GitHub Repository: [Research Materials](#)

PRESENTATIONS

Yan, M., Dangeti, R., Meng, A., Zhu, D., Eiden, E., King, C., Kannampallil, T., & Abraham, J.

Iterative User-Centered Evaluation of a Machine-Learning Augmented Postoperative Risk Summary Report

Poster presented at the Biomedical Informatics and Data Science (BIDS@I2DB) poster session, Washington University School of Medicine, July 2026.

PROFESSIONAL EXPERIENCE

Wellesley College Clapp Library, Wellesley, MA

Sep 2023 - Present

Knapp Makerspace & Interactive Media Lab

Fabrication Consultant

- Lead hands-on workshops in textile arts and digital fabrication to foster an inclusive maker community, and design promotional materials using Photoshop and Canva to increase engagement.
- Provide one-on-one training on several fabrication machines, including sewing machines, embroidery tools, and 3D printers, adapting instruction for users of all skill levels.

- Troubleshoot and maintain Prusa 3D printers, diagnosing software and hardware issues to ensure smooth operation for student projects.

Handshake AI, Remote

Nov 2025 - Mar 2026

General AI Trainer (Visual & Design-Focus), MOVE Fellow

- Conducted multimodal training and evaluation of large language models (LLMs) across text, audio, and image tasks to improve model performance, safety, and output quality.
- Trained and evaluated AI systems on Blender animations and Inkscape vector designs to improve model understanding of graphics, motion, and visual composition.

GoInvo, Arlington, MA

Jun 2025 - Aug 2025

Healthcare Design Intern

- Built an open-source interactive website using Figma and GitHub Pages to visualize eight state policy models for psychedelic therapy legalization by putting together clinical research, state reports, and policy data into accessible user flows that highlight psilocybin's potential in treating mental illnesses such as PTSD and depression.
- Collaborated with a team of designers and a State Public Health Service to build a website showcasing dashboards that visualize state-level public health metrics.
- Rapidly prototyped interactive data visualizations in Figma, Cursor, and Node.js; received feedback from project leads and teammates and iteratively refined the designs for policy-making and educational use.

Aveyo Avocado Foods, Boston, MA

May 2024 - Aug 2024

Graphic Design & Brand Management Intern

- Collaborated with the Founder/CEO on brand management and website development for a new product line, building the site architecture (sitemapping, wireframing) and leading a website redesign with custom visual assets, contributed to a launch across 145 Whole Foods stores nationwide.
- Conducted research across condiment and healthy-eating brands to inform packaging design, brand positioning, and a multi-month social media strategy, and supported investor communications with a redesign of the investor pitchdeck.

HONORS & AWARDS

- Selected Participant of Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS) 2026, a one-week international research school for pre-doctoral scholars ~12% acceptance rate. One week, fully-funded program hosted by the Max Planck Institute for Software Systems with Cornell University and the University of Maryland, featuring lectures and mentoring from leading computer science researchers.
- 2023 National Scholastic Art & Writing Gold Medal, Illustration category (2 pieces)

SKILLS

Technical: HTML/CSS/JavaScript, Node.js, Python, Java, C#, Plotly Dash, Git/Github, Adobe Creative Suite, Figma, Blender, Spline 3D, Autodesk Fusion, Shapr3D, SketchUp

Languages: Chinese (Bilingual), Spanish (Proficient)

ACTIVITIES

Wellesley in Product, MIT Club Wrestling, Wellesley College NCAA DIII Cross Country and Track & Field (formerly), Wellesley College NCAA DIII Crew (formerly)

REFERENCES

Available upon request.